SARDAR PATEL INSTITUTE OF TECHNOLOGY
(Autonomous institute affiliated to Mumbai University)
MUNSHI NAGAR, ANDHERI W, MUMBAI – 400058.

CRITERION VII

INSTITUTIONAL VALUES AND BEST PRACTICES

7.1.7 DISABLED FRIENDLY

INDEX

Bhartiya Vidya Bhavans
Sardar Patel Institute of Technology
(An Autonomous Institute Affiliated to University of Mumbai)
Munshi Nagar, Andheri (W), Mumbai 400058

List of Sponsored Research Projects for the AY 2018-19 undertaken by the Faculty under the category of Disability

Sr no.	Academic Year	Department	Funding Agency	Name of the Faculty	Project Title	Duration/Research Grant Amount (Rs)
1	2019-20	MCA	Minor Research Grant University of Mumbai	Prof. Pallavi Thakur	Helping Hand	1 Year/25000

List of Mini and Major Projects for the AY 2018-19 to 2022-23 undertaken under the category of Disability

Sr no.	Academic Year	Branch	Subject code	Name of the students	Project Title	Project Guide
1	2018-19	ETRX	ETP81	Joshi Himanshu Devvrat Mehta Darshana Deepak Nerkar Mrunal Dattatray	Sign Speak Gloves	Prof. Priya Despande
2	2018-19	ETRX	ETP81	Mestri Rohan Satyavan Khuteta Sagar Sabjay Prathamesh Limaye Vivek	Lip Reading using Deep Learning	Prof. M. R. Bansode
3	2018-19	ETRX	ETP81	Joshi Himanshu Devvrat Mehta Darshana Deepak Nerkar Mrunal Dattatray	Sign Speak Gloves	Prof. Priya Despande
4	2019-20	IT	ITP71/81	Riya Deshpande Vishakha kalal Prathmesh Patkar	Depression Detection System	Prof. Rupali Sawant
5	2019-20	IT	ITP71/81	Pradnya Banate Gaurav Gosavi Chetna Chaudhari	Prediction of mental health illness	Prof. Rupali Sawant

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Sr no.	Academic Year	Branch	Subject code	Name of the students	Project Title	Project Guide
6	2020-21	IT	ITP71/81	Shubham Gogate Atul Kamble Dhawal Mali	Hand Gestures Recognition using Inertial Sensors through Deep Learning	Prof. Jignesh Sisodia
7	2020-21	ETRX	ELP65	Lance Fernandes Prathamesh Dalvi Akash Junnarkar	Sign Language Translator	Prof. Manisha Bansode
8			ELP65	Janhavi Bhopale Atharv Mane Ria Motghare	Speech Cognizance and Transcription	Prof. Priya Despande
9	2021-22	ETRX	ELP71	Isha Naik Trisha Patel Divya Raut	Prosthetic Arm	Dr.Sutar
10			ET208	Shreyash Kokate Rahul,Joshi Abhinav Adhole	Robotic Arm	Prof. Najib Ghatte
11	2022-23	EXTC	EC208	Aditya Rane Swapnil Sawant Shubham Surve	Sign to voive converting glove for mute people	Prof. Manish Parmar
12			EC208	Henil Gandhi Pratham Gajapure Nishant Chandevila	Smart stick for blind people for step and location tracking	Prof. Priya Despande
13			EC208	Kunuj Chauhan Sankalp Dayal	Wearable smart glove for speech impared	Prof. Manisha Bansode

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Helping Hand: Metro Travel Assistance App for Elderly and Differently Abled People

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Abstract— **As per Census 2011 there are 59,392 persons with disabilities in Maharashtra and it is 2.63 % of total population of Maharashtra State. An aging population increases the demand for health services and assistance services. The problem of assisting and helping in day to day needs for the elderly has increasingly become one of the focuses of social concerns. Because of the popularization of mobile devices and wearable devices like smartwatches, using these devices to assist the care of the elderly has become a feasible solution. Differently abled or elderly people can navigate through the places that they have already been in with rather ease. But if they have to go to a new, unfamiliar place, that is where they face some issues. In this paper, we developed an elderly assistant system based on mobile and wearable devices, which includes two apps i.e., the Guardian side and the Elderly side. The guardian side remotely helps the elderly in case of emergency SOS request; the Elderly side assists the traveling needs and emergency needs of the elderly/differently abled people. The Developed assistance app provides real time travel assistance in metro stations, and SOS support in emergency situations. The aim of this assistance app is to provide a trustworthy system where the needy can travel safely and get the required assistance.**

Keywords— mobile, elderly, public transport, helping, travel

I. INTRODUCTION

Public transport is a blessing for the masses who have to commute frequently, especially in a crowded city like Mumbai. But the same public transport becomes a maze for the elderly and differently abled people, who are helpless in the fast-paced life of a metropolitan city. The majority of the differently abled rely on public transport due to their financial inability to have their own vehicles with chauffeur facility and also due to their inability to operate the vehicles on their own. Even though the infrastructure exists to provide assistance to those in need, in these public transport stations, it is either not known or is a big hassle for elderly and differently abled persons to use. The result of this is a poor transport system, for those who are either elderly or differently abled, to navigate to their destination.

Problem we have with public transport is not the unavailability of the system to help such people, but the method to leverage these facilities in day-to-day use. The system is unnecessarily complex and lacks the basic features of a service like accountability, responsivity, responsibility, etc. An elderly traveler in this public transport faces a major problem of tracking the changes timings of trains.

In this proposed solution, we introduce an ecosystem of mobile applications which aims to solve all the above-mentioned problems. We are proposing a system of 3 mobile applications which will work in sync to provide the user with all necessary matters like searching for metro, requesting assistance on the station and raising a SOS request in case of an emergency.

II. RELATED WORK

According to a research, titled Situation Analysis Of The Elderly in India[1], The population of senior citizens (aged 60 or above) accounts for 7.4% of total population as of 2001. Out of this population, around 65 percent are dependent on someone else for their daily chores. According to the World Health Organization (WHO), 10% of the population in developed countries and 12% of the population in developing countries are especially abled people. In our country, disabled people face a lot of difficulty in daily outdoor activities and traveling despite the existence of special rules and regulations for them. The solution can be redesign of buildings, transportation systems and city life to make travel easier for differently abled people.

The same observations are supported in another research paper by the name Understanding the travel behavior of elderly people in the developing country: a case study of Changchun, China[2] stated that After World War II, the Baby-Boom phenomena and decline in the number of deaths of elderly, has resulted in an increase in elderly population. Around 91.59% of elderly are retired and 98.02% of them do not own a car. Hence, the usage of motor cars is less in elderly. We can find that both the elderly males and females choose walking and public transport as their main travel mode. Approximately more than 43% of elders use public mode of transport and less than 1% of them have motor cars. Male older people's average number of trips per day is 0.108 trips per day higher than females.

As observed from above references[1][2] that the number of senior citizens is increasing with every passing year and thus a system which can assist the elderly to help them in transit should be considered as a priority.

A study published in 2015 titled Review of Public Transport Needs of Older People in European Context[3] states that as people get older, they might become unfit to drive due to many issues. Even if they are fit, there is a lack of sensitivity towards older drivers. This is why most of them prefer public transport Hence, public transport is important to older people's quality of life, their sense of freedom and independence. The requirements of older people using public transport were studied in terms of four main issues: Affordability, availability, accessibility and acceptability. Increased traffic density and flow, as well as some drivers' lack of consideration for other road users. These characteristics tend to affect older people more than other age groups. As a consequence, older people tend to make fewer journeys than other adults and may change their transport mode.

According to Impacts of COVID-19 on access to transportation for people with disabilities[4], People with disabilities have several obstacles when it comes to using transportation to acquire vital commodities and services, such as fresh food and medical treatment, that are necessary for preserving health, and the epidemic is exacerbating many of these challenges. difficulties getting needed assistance using transportation and completing activities of daily living ranging from personal care to getting groceries, and difficulties getting needed assistance using transportation and completing activities of daily living ranging from personal care to getting groceries.

Same observations are given by a paper titled Traveling With a Disability More Than an Access Issue[5], they face many practical and social obstacles that can inhibit their full participation in tourism. People with disabilities have more things to consider and more challenges to face before and during a trip than those without. It provides insight into the tourism experiences of people with disabilities, in particular those with mobility or visual impairment. With better understanding of their experiences, it is hoped that the society at large will be more aware of their needs.

Supported by the previously mentioned points in reference[2][3][4][5], we can observe that the number of elderly/disabled people are traveling more in public transport. The nature of travel is not recurrent so they are unaware of the surroundings and would need assistance physically.

As stated in a paper titled Assistive technology and passengers with special assistance needs in air transport: contributions to cabin design[6], the word assistive technology is used for a variety of devices or services that are responsible for helping people with special needs, thereby trying to eliminate the factor of activity limitation and restriction for social participation.

As mentioned in proposed services by Mumbai Metro titled Mumbai Metro Transforming Transport[7], the Mumbai Metro will also provide several facilities to the differently abled and improve accessibility for them. It is planned to have allocated space for wheelchairs in trains, platform-level boarding, elevators, escalators, accessibility ramps, allocated waiting areas at each station, dedicated seats for the physically disabled passengers in each coach, and priority e-ticketing counters, among other facilities. From a special needs point of view, 3% of the MMR population is differently abled. The lack of escalators, elevators, accessibility ramps, and platform-level boarding makes commuting difficult for this group and, hence limits their development and opportunities.

Same observation can be seen in another research titled Examination of accessibility for disabled people at metro stations[8], the platforms should be designed in a way that also makes it easy for wheelchair users to easily move around. The safety lines on the platform should have proper colors, textures and patterns so it can be easily seen by people with visual impairment problems.

As per observations in reference[7][8], we can conclude that an additional support of devices and infrastructure would not only elders/disabled people to ease the hassle of traveling but also encourage them to travel on their own resulting in self-reliance for them. So, in our proposed solution we have introduced use of a wearable device to make the experience of the users more comfortable.

As per above observations, we can conclude that the metro has an existing infrastructure to help disabled/elderly people but inculcating our proposed solution would result in accountability, reliability on services, transparency in assistance given. This in turn would support more infrastructural growth of this kind of system.

III. SYSTEM STRUCTURE AND FUNCTIONS

As mentioned in the introduction, we developed three mobile applications, Helping Hand, Helping Hand Admin and Helping Hand Guardian. Helping Hand application which would be used by the users to avail the services, Admin application would be used to validate and deploy assistance and the Guardian application would be used to notify the guardian in case any immediate assistance is needed.

Following is the list of the applications and the users who will use them;

- Helping Hand – Users (Differently Abled People, Elderly People)
- Helping Hand Admin – Station Administrator, On Ground Support Staff
- Helping Hand Guardian – Guardian of user

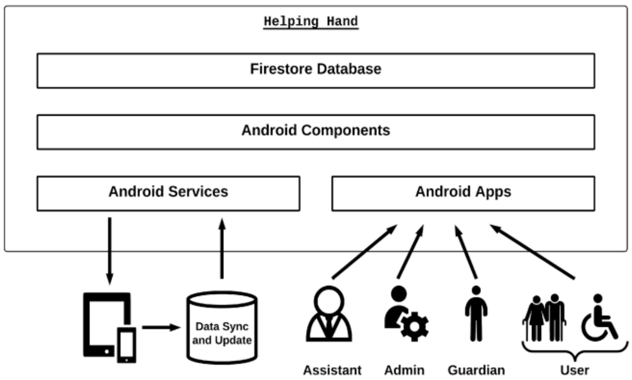

Fig. 1: Architectural design

Registration has the following flow in the proposed ecosystem of the applications;

Differently abled or Elderly people can register themselves on the application by providing the mobile number and then validating it by entering the received One Time Password. After this a profile screen is displayed to the user to create their profile where they have to provide profile photo, name, disability/es, address, guardian name, guardian contact details. After providing all the details the user will be registered on the system.

Station Administrator is registered by the application administrator in the backend and credentials are given to them. Station Administrators can then register On Ground Support Staff for their respective stations by entering the contact number of the support staff on the admin application and then validating it by entering the One Time Password received by the staff. After adding the basic details like employee id, employee name, profile photo and Aadhaar number, ground staff will be added.

Guardian enters their phone number in the Guardian application. System checks if the entered number is listed alongside any existing user. If the condition is satisfied, a one-time password is sent to the guardian for authentication. After successful authentication, the user is prompted to scan a QR code, which can be found in the user application's profile section. Once scanned, the user is mapped with the guardian and guardian is onboarded.

Following are the major modules in the proposed solution;

A. Searching for train

The user application is specifically designed for the people who are elderly and face problems while using modern day technologies. Interactive elements have been sized to a view, which can help those who have a poor vision. There are different ways of searching trains after entering the source and destination stations. Users can choose to view the next three trains available which are displayed on the screen by default. Users can press next or previous buttons to view the next three or previous three trains timings respectively. Users can also view all the available trains throughout the day by pressing the "View All Trains" button. Apart from this, users can search trains by entering a specific time of the day.

B. Raising assistance request

During their journey, users may face difficulties because of their physical limitations. To help users, Metro appoints assistants who provide all the help users need for their journey. To request assistance, the user presses the "Request Assistance" button. This navigates the user to a page where the user is required to select the station he is currently present at and allow location permissions to the app in order to successfully raise an assistance request. The user then waits until an assistant is assigned to them.

C. Resolving assistance request

For a single admin there can be many requests coming at the same time. So, to solve this problem there will be individual admins at every station. These admins will be responsible for looking after the day-to-day administrative tasks of the station. Whenever a user sends an assistance request, a notification pops up on the respective station admin's application who then checks the legitimacy of the request. The assistance request consists of date and time when it was raised, profile of the user who raised it and their current location. The admin checks the location and confirms if it is in the vicinity of the metro station. If the user is found to be in the nearby area, the admin accepts the request, otherwise the request is rejected.

The assignment of assistants is decided by our algorithm that checks the number of assistants that are currently online and the number of people that the individual assistants have helped throughout the day. The assistant who is online and has the least count of completed requests, is assigned the current request.

When the assistant is assigned to the user, the assistant's details can be viewed on their screen. Similarly, the assistant can also now view the details of the user assigned along with profile photo and his location.

D. Raising SOS request

The user will have a wearable device on them all the time, which in case of emergency raises a SOS alarm. This SOS message containing the location of the user will be sent to the guardian. The guardian application running in background will be having an active thread to check for the sender of the received message and check if the message was sent by the user wearing the device. Once the sender is determined to be the user, the guardian application will read the content of the SMS and relay the same to the Administrator of the station after determining the metro station closest to the user's location. This will raise a notification on the admin application. The admin can view the content of the SOS request which includes user details like name, contact number of the user and their disability, SOS details like address, SOS location link, date and time of the SOS raised.

E. Resolving SOS request

The admin will click on the SOS location link from the received SOS request. By doing so the app will redirect the admin to a browser window in which the link will be opened. The admin can then view and verify the location of the user. If the user is in the vicinity of the metro station, then the admin can deploy the emergency services to the user's location.

IV. RESULT AND DISCUSSION

The developed system is a simple widget-based application capable of providing options to search trains, raising assistance requests, to view schedules of the trains and communicate with Metro officials in case of emergency with the help of guardian app. Application is working as intended and sends a message every time after pressing the SOS button. Implementation of the application shows that, assistant response time is not more than a few minutes for every request raised. It solves safety and emergency issues for elderly and differently abled people who can be in life threatening or suspicious situations.

Fig 2 shows the Search Train Screen which also has the Request Assistant Screen of the user app where they would be required to select the station when needing assistance.

After a request is processed from the user's end, it is forwarded to the admin for approval. The admin verifies the location and then approves or rejects the request based on the user's location and data as shown in Fig 3.

The allotted assistant is shown on the user's application as shown in Fig 4.

In case of an emergency, once an SOS is triggered from the wearable device of the user, the SOS is forwarded to the Guardian via SMS as shown in Fig 5. The guardian then relays the SOS information to the admin of a specific station.

The admin after receiving an SOS request gets a notification like Fig 6. The admin then can view the SOS request in the SOS section of the admin application. After the admin validates the request, they can deploy the help by clicking the button ‘Deploy Help’ as shown in Fig 7.

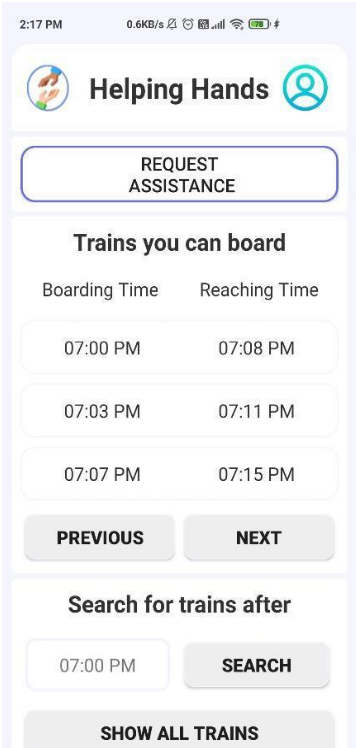

Fig. 2: Landing Screen of User’s app

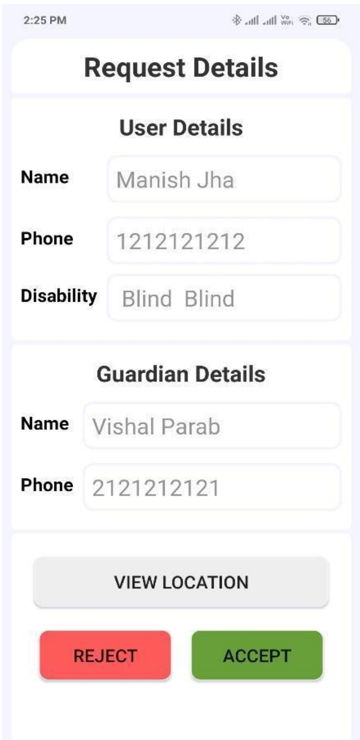

Fig. 3: Request Approval Screen of Admin’s app

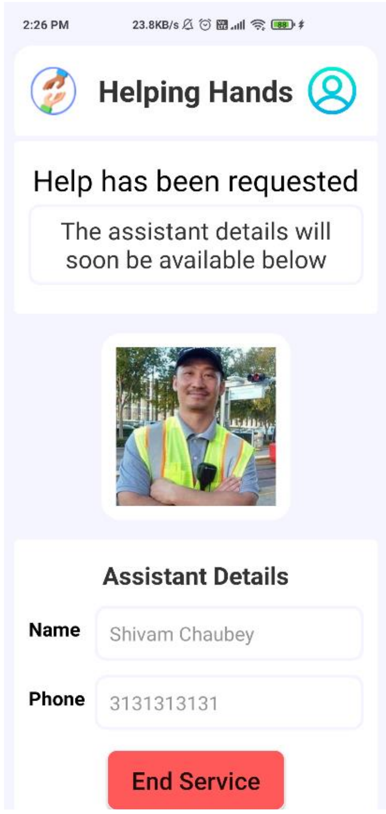

Fig. 4: View assistant screen of User’s app

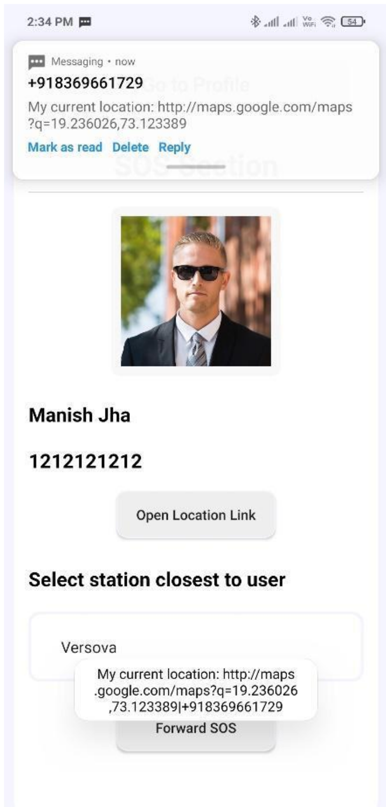

Fig. 5: View SOS screen of Guardian’s app

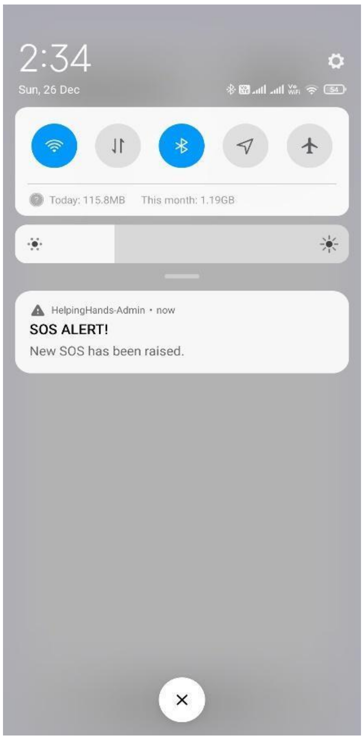

Fig. 6: SOS alert for admin

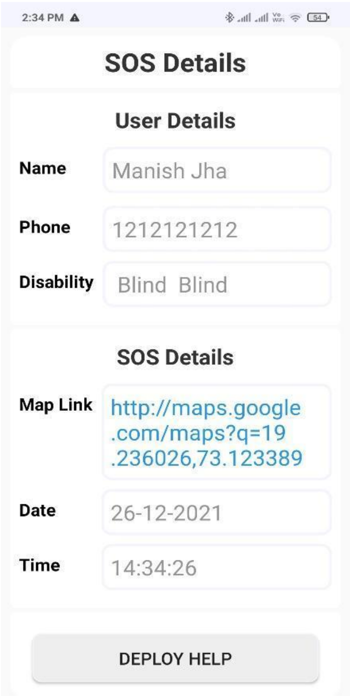

Fig. 7: SOS alert for admin

V. FUTURE ENHANCEMENT

The developed system is a simple widget-based application capable of providing options to search trains, raise assistance requests, view schedules of the trains, and communicate with Metro officials in case of emergency with the help of the guardian app. The application is working as intended and sends a message every time after pressing the SOS button. Implementation of the application shows that assistant response time is not more than a few minutes for every request raised. It solves safety and emergency issues for elderly and differently-abled people who can be in life-threatening or suspicious situations.

Even though most people in India can read common English instructions and operate an app, some people may prefer using the app in their native languages. These include majorly spoken languages like Marathi, Hindi, Tamil, Telugu, Bengali, Gujrati and other regional languages. This can make the app more usable for a variety of users and improve the overall user experience.

Currently the app depends on a third-party wearable device to send SOS messages. The guardian app detects when a SOS message is received and then lets the guardian decide to which station admin the SOS is supposed to be forwarded. Having a custom designed wearable device enables us to make the flow concise and much faster. The custom device can then directly send the SOS message to the respective station admin without requiring intermediate processing by guardian app.

The SOS flow can be improved by eliminating the need of a relay app i.e., the Guardian app. The latitude and longitude of the user can be pinpointed using a location API on the User app whenever a SOS message is sent. This enables the user app to directly relay the SOS to the respective station admin without need of the guardian to forward it. This change can make the flow more reliable by reducing the number of hops required for the message to reach its destination thus making the process faster.

VI. CONCLUSION

Differently abled and senior citizens have a lot of trouble when trying to travel through crowded places like metro stations. Even though there are many rules and regulations to support them, they are not always comfortable traveling alone and mostly need help to perform activities like buying tickets, climbing stairs or escalators, getting into lifts, boarding the train, etc. The metro staff always tries to help such people but there must be some kind of link or communication between the staff and the disabled/elderly people. Having such a link enables the disabled/elderly people to communicate with the staff for help. The Helping Hand app acts as the link between the two parties. The app lets users view all the train timings and request assistance from the metro staff. The app also keeps a watch on the SOS system of the user’s wearable device so it can forward the SOS to the metro staff. Having such an app means the disabled/elderly people don't have to seek for help but just ask for it which makes their traveling experience a lot easier and more comfortable.

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